

COM3067 – ALGORITHMS

Vize ve Final Çalışma Soruları

Önemli Not: Bu dokümanda yer alan soruların tamamı, *The Design and Analysis of Algorithms* (Anany Levitin) adlı ders kitabı ve ilgili dersin geçmiş yıllarda yapılmış sınavları referans alınarak hazırlanmıştır.

EFFICIENCY ANALYSIS & BRUTE FORCE & EXHAUSTIVE SEARCH

1. What value is returned by the following algorithm? What is its basic operation? How many times is the basic operation executed? Give the worst-case running time of the algorithm using Big Oh notation.

CRYOWAR(n)

input: an integer n

$r \leftarrow 0$

for i = 1 to n

for j = i to n

for k = j to i + j

$r \leftarrow r + 1$

return r

2. What does the following algorithm compute? What is its basic operation? How many times is the basic operation executed? Give the worst-case running time of the algorithm using Big Oh notation.

ALASKA(A = (a_{ij})_{n×n})

input: an $n \times n$ matrix of real numbers

$r \leftarrow 0$

for i = 1 to n-2

for j = i + 1 to n

if a_{jj} != a_{ji}

return false

return true

3. List the following functions according to their order of growth from the lowest to the highest.

$$(n-2)!, 5 \lg(n+100)^{10}, 2^{2n}, 0.001n^4 + 3n^3 + 1, \ln^2 n, \sqrt[3]{n}, 3^n.$$

4. Order the following functions according to their order of growth (from the lowest to the highest). If any two or more are of the same order, indicate which.

$$f_1(n) = n^2 + \log n$$

$$f_8(n) = n^{12} + n^{10}$$

$$f_2(n) = \sqrt{n}$$

$$f_9(n) = n^{12} \log n$$

$$f_3(n) = n - 1000$$

$$f_{10}(n) = n^{\frac{1}{3}} + \log n$$

$$f_4(n) = n \log n$$

$$f_{11}(n) = (\log n)^2$$

$$f_5(n) = 2^n + n^{10}$$

$$f_{12}(n) = 10^{15}$$

$$f_6(n) = n^5 + 3^n$$

$$f_{13}(n) = \frac{n}{\log n}$$

$$f_7(n) = n^{11} \cdot 2^{2 \log n}$$

$$f_{14}(n) = \log \log n$$

5. Devise a brute-force algorithm that takes a string of n characters as input, and outputs the number of palindromic substrings in the given string.

6. There is a sequence of n tasks $[t_1, t_2, \dots, t_n]$ where every task has a deadline d_i and an associated profit p_i . Every task takes a single unit of time and only one task can be scheduled at a time. Also, the profit p_i can only be earned if the associated task is finished before the deadline d_i . Under these constraints, devise a brute-force algorithm that takes such a sequence as input, and outputs a scheduling of tasks having the maximum possible total profit.

7. There are n stacks of n identical-looking coins. All of the coins in one of these stacks are counterfeit, while all the coins in the other stacks are genuine. Every genuine coin weighs 10 grams; every fake weighs 11 grams. You have an analytical scale that can determine the exact weight of any number of coins.

- (a) Devise a brute-force algorithm to identify the stack with the fake coins and determine its worst-case efficiency class.
- (b) What is the minimum number of weighings needed to identify the stack with the fake coins?

8. A firm wants to determine the highest floor of its n -story headquarters from which a gadget can fall without breaking. The firm has two identical gadgets to experiment with. If one of them gets broken, it cannot be repaired, and the experiment will have to be completed with the remaining gadget. Design an algorithm in the best efficiency class you can to solve this problem.

9. Consider the classic puzzle of placing eight queens on an 8×8 chessboard so that no two queens are in the same row or in the same column or on the same diagonal. How many different positions are there so that

- (a) no two queens are on the same square?
- (b) no two queens are in the same row?
- (c) no two queens are in the same row or in the same column?

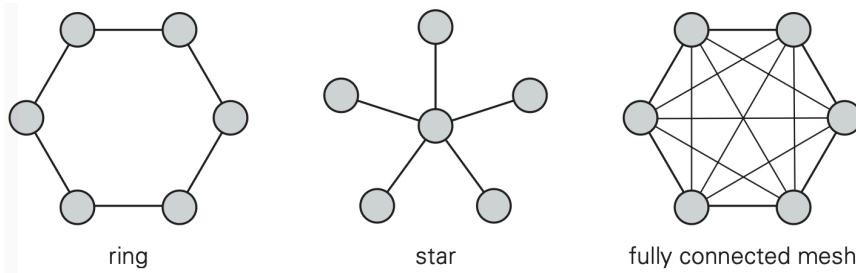
Also estimate how long it would take to find all the solutions to the problem by exhaustive search based on each of these approaches on a computer capable of checking 10 billion positions per second.

10. You have a row of $2n$ disks of two colors, n dark and n light. They alternate: dark, light, dark, light, and so on. You want to get all the dark disks to the right-hand end, and all the light disks to the left-hand end. The only moves you are allowed to make are those that interchange the positions of two neighboring disks.



Design an algorithm for solving this puzzle and determine the number of moves it takes.

11. A network topology specifies how computers, printers, and other devices are connected over a network. The figure below illustrates three common topologies of networks: the ring, the star, and the fully connected mesh.



You are given a boolean matrix $A[0 \dots n-1, 0 \dots n-1]$, where $n > 3$, which is supposed to be the adjacency matrix of a graph modeling a network with one of these topologies. Your task is to determine which of these three topologies, if any, the matrix represents. Design a brute-force algorithm for this task and indicate its time efficiency class.

12. Determine the number of character comparisons made by the brute-force algorithm in searching for the pattern "GANDHI" in the text "THERE_IS_MORE_TO_LIFE_THAN_INCREASING_ITS_SPEED". Assume that the length of the text—it is 47 characters long—is known before the search starts.

GRAPH THEORY & BFS/DFS

1. A graph has been explored by a depth-first search (DFS) algorithm; and the discovery and finishing times of its vertices are given as A(1, 14), B(3, 4), C(10, 13), D(2, 5), E(6, 9), F(7, 8), and G(11, 12). Using only this information,

- (a) Write down a possible graph.
- (b) Construct the adjacency list for the possible graph.
- (c) Construct the adjacency matrix for the possible graph.

2. A bipartite graph is a graph whose vertices can be divided into disjoint subsets U and V such that every edge in the graph connects a vertex in U to a vertex in V . Devise an algorithm that uses breadth-first search (BFS) or depth-first search (DFS) as a sub-procedure, takes an $n \times n$ adjacency matrix of a graph as input, and checks whether the graph is bipartite or not.

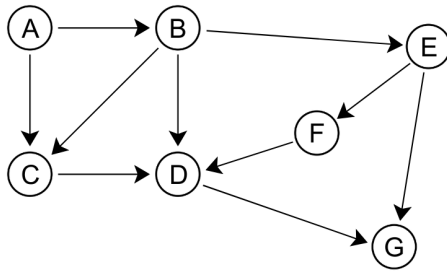
3. One can model a maze by having a vertex for a starting point, a finishing point, dead ends, and all the points in the maze where more than one path can be taken, and then connecting the vertices according to the paths in the maze.



Which traversal—DFS or BFS—would you use if you found yourself in a maze and why?

4. Devise an algorithm that takes an $n \times n$ adjacency matrix of a directed graph as input, and determines whether the directed graph contains a simple cycle of odd length or not. The running time of your algorithm should be $O(n + m)$ where n is the number of vertices and m is the number of edges contained in the graph.

5. Consider the following graph.

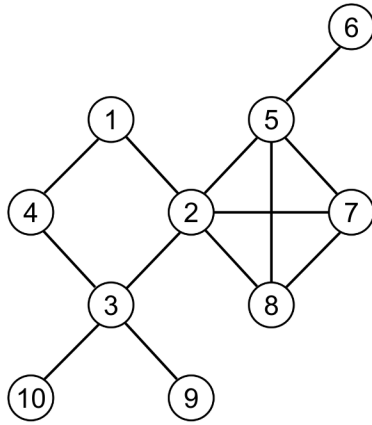


(a) Show the adjacency list of the graph.

(b) Apply breadth-first search (BFS) algorithm to the graph using the vertex a as the source vertex. For the answer, just draw the BFS tree that you obtain after the algorithm terminates (to get full credit, you should write the final keys near to the corresponding vertices in the graph as shown in the slides).

(c) Apply depth-first search (DFS) algorithm to the graph using the vertex a as the source vertex. For the answer, just draw the DFS tree that you obtain after the algorithm terminates (to get full credit, you should write the discovery-finish times of the vertices near to the corresponding vertices in the graph as shown in the slides).

6. Consider the following graph.

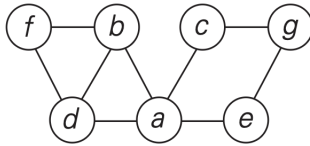


(a) Show the adjacency list of the graph.

(b) Apply breadth-first search (BFS) algorithm to the graph using the vertex a as the source vertex. For the answer, just draw the BFS tree that you obtain after the algorithm terminates (to get full credit, you should write the final keys near to the corresponding vertices in the graph as shown in the slides).

(c) Apply depth-first search (DFS) algorithm to the graph using the vertex a as the source vertex. For the answer, just draw the DFS tree that you obtain after the algorithm terminates (to get full credit, you should write the discovery-finish times of the vertices near to the corresponding vertices in the graph as shown in the slides).

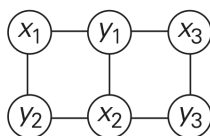
7. Consider the following graph:



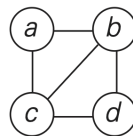
(a) Write down the adjacency matrix and adjacency lists specifying this graph. (Assume that the matrix rows and columns and vertices in the adjacency lists follow in the alphabetical order of the vertex labels.)

(b) Starting at vertex a and resolving ties by the vertex alphabetical order, traverse the graph by depth-first search and construct the corresponding depth-first search tree. Give the order in which the vertices were reached for the first time (pushed onto the traversal stack) and the order in which the vertices became dead ends (popped off the stack).

8. A graph is said to be *bipartite* if all its vertices can be partitioned into two disjoint subsets X and Y so that every edge connects a vertex in X with a vertex in Y. (One can also say that a graph is bipartite if its vertices can be colored in two colors so that every edge has its vertices colored in different colors; such graphs are also called *2-colorable*.) For example, graph (i) is bipartite while graph (ii) is not.



(i)



(ii)

(a) Design a DFS-based algorithm for checking whether a graph is *bipartite*.

(b) Design a BFS-based algorithm for checking whether a graph is *bipartite*.

RECURRENCE RELATION & MASTER THEOREM

1. Solve the following recurrence relation using the Master Theorem.

$$T(n) = \begin{cases} 1, & n = 0 \\ 5T(n/2) + n^2, & n > 0 \end{cases}$$

2. Solve the following recurrence relation using Master Theorem.

$$T(n) = \begin{cases} 1, & \text{if } n = 1 \\ 9T\left(\frac{n}{3}\right) + n^2, & \text{if } n > 1 \end{cases}$$

3. Solve the following recurrence relation using Master Theorem.

$$T(n) = \begin{cases} 1 & , \quad \text{if } n \leq 2 \\ 2T\left(\frac{n}{2}\right) + n \log n, & \text{if } n > 2 \end{cases}$$

4. Solve the following recurrence relation using recursion tree method.

$$T(n) = \begin{cases} 1 & , \quad \text{if } n \leq 2 \\ T\left(\frac{n}{3}\right) + T\left(\frac{2n}{3}\right) + n, & \text{if } n > 2 \end{cases}$$

5. Solve the following recurrence relation using Master Theorem.

$$T(n) = \begin{cases} 1, & \text{if } n = 1 \\ 7T\left(\frac{n}{3}\right) + n^2, & \text{if } n > 1 \end{cases}$$

6. What does the following recursive algorithm compute? Set up a recurrence relation for the running time of the algorithm and solve it using backward substitution.

SAMSUN($a_i, \dots, a_j$)

input: a sequence of integers

if $i = j$

return a_i

else

$\text{mid} \leftarrow \text{floor}((i + j) / 2)$

$\text{temp1} \leftarrow \text{SAMSUN}(a_i, \dots, a_{\text{mid}})$

$\text{temp2} \leftarrow \text{SAMSUN}(a_{\text{mid}+1}, \dots, a_j)$

if $\text{temp1} \leq \text{temp2}$

return temp1

else

return temp2

DECREASE AND CONQUER

1. Given n coins, all the same except for one fake coin is lighter than others, and a balance scale allowing you to compare any two piles of coins, devise an algorithm that finds the lighter coin with the minimum possible number of weights. The running time of your algorithm should be better than $O(\log n)$.

2. Given two arrays of lengths n , a conceptual two-dimensional $n \times n$ array is formed whose rows and columns are sorted in increasing order. An integer x is also provided. The following brute-force algorithm is employed:

BRUTE_FORCE(A, x)

input: An $n \times n$ matrix A and an integer x

```

for  $i \leftarrow 0$  to  $n - 1$ 
  for  $j \leftarrow 0$  to  $n - 1$ 
    if  $A[i, j] = x$  then
      return true
return false

```

(a) Briefly explain what this algorithm does.

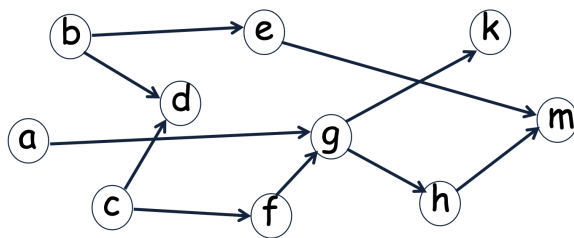
(b) Design a decrease-and-conquer algorithm that performs the same task in $O(n)$ running time.

3. There are n pancakes of all different sizes that are stacked on top of each other. You are allowed to slip a flipper under one of the pancakes and flip over the whole stack above the flipper. The purpose is to arrange pancakes according to their size with the biggest at the bottom. Design an algorithm that runs in $O(n^2)$ time.

4. Given a sorted sequence of n items $[a_1, a_2, \dots, a_n]$ and a search key K . Devise a decrease-and-conquer algorithm that determines whether the sorted sequence contains the key or not.

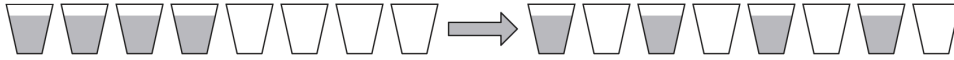
5. A celebrity among a group of n people is a person who knows nobody but is known by everybody else. Design a decrease-and-conquer algorithm that identifies a celebrity among n people, or outputs “no celebrity identified” if there is no such person, by only asking the question "Do you know him/her?" to people. Analyze the efficiency of the algorithm.

6. Consider the following directed graph:



Apply topological sort and write at least two correct outputs upon the termination of the algorithm.

7.



- (a) There are $2n$ glasses standing next to each other in a row, the first n of them filled with a soda drink and the remaining n glasses empty. Make the glasses alternate in a filled-empty-filled-empty pattern in the minimum number of glass moves.
- (b) Solve the same problem if $2n$ glasses— n with a drink and n empty—are initially in a random order.

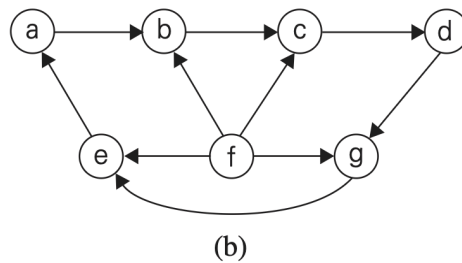
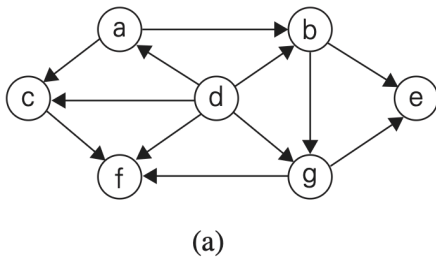
8. Apply insertion sort to sort the list E, X, A, M, P, L, E in alphabetical order.

9. Devise an algorithm that takes a sorted sequence of n distinct integers $a_1, \dots, a_n$ as input, and outputs 'YES' if there is an index i such that $a_i = i$ and 'NO' if there is no such index. The running of your algorithm should be at most $O(\log n)$.

10. Devise an $O(n)$ -time algorithm that takes an array of length n , which consists of the letters B , M , and W , and sorts them in a way that all the B s come before all the M s, and all the M s come before all the W s.

11. You have the results of a completed round-robin tournament in which n teams played each other once. Each game ended either with a victory for one of the teams or with a tie. Design an algorithm that lists the teams in a sequence so that every team did not lose the game with the team listed immediately after it. What is the time efficiency class of your algorithm?

12. Apply the source-removal algorithm (topological sorting) to the digraphs given below.



13. Apply quickselect to find the median of the list of numbers given below.

9, 12, 5, 17, 20, 30, 8.

14. You need to search for a given number in an $n \times n$ matrix in which every row and every column is sorted in increasing order. Design a decrease-and-conquer algorithm for this problem. The running time of your algorithm should be $O(n)$.

DIVIDE AND CONQUER

1. Given two sorted arrays of equal size n , find the median of the union of the two arrays in $O(\log n)$ time using a divide and conquer approach.

2. Apply mergesort to sort the list E, X, A, M, P, L, E in alphabetical order.

3. Compute 2101×1130 by applying a divide-and-conquer algorithm.

4. Devise an $O(n \log n)$ -time divide-and-conquer algorithm that takes a sequence of n integers, outputs the maximum sum of any contiguous subsequence. Also, all the inputs given to the algorithm are assumed to be legitimate, i.e. the users are assumed to provide only integers.

5. Design a divide-and-conquer algorithm for computing the number of levels in a binary tree. (In particular, the algorithm must return 0 and 1 for the empty and single-node trees, respectively.) Discuss the running time of your algorithm.

6. Consider the matrices $A_0, A_1, A_2, \dots$ that are recursively formed as follows:

$$A_0 = [1]_{1 \times 1}, \text{ and } A_1 = \begin{bmatrix} A_0 & A_0 \\ A_0 & -A_0 \end{bmatrix}_{2 \times 2} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}_{2 \times 2}$$

$$A_k = \begin{bmatrix} A_{k-1} & A_{k-1} \\ A_{k-1} & -A_{k-1} \end{bmatrix}_{2^k \times 2^k} \text{ where } A_{k-1} \text{ is the } 2^{k-1} \times 2^{k-1} \text{ matrix.}$$

Devise a divide-and-conquer algorithm that takes a column integer vector v whose length is $n = 2^k$, and computes the matrix-vector product $A_k * v$. The running of your algorithm should be at most $O(n \log n)$.

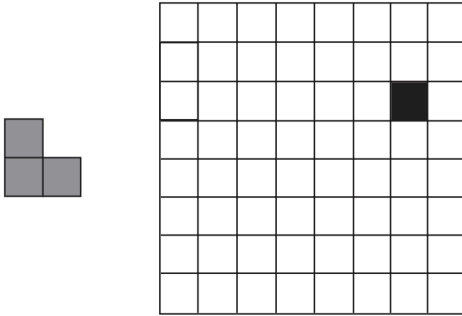
7. Apply Strassen's algorithm to compute

$$\begin{bmatrix} 1 & 0 & 2 & 1 \\ 4 & 1 & 1 & 0 \\ 0 & 1 & 3 & 0 \\ 5 & 0 & 2 & 1 \end{bmatrix} * \begin{bmatrix} 0 & 1 & 0 & 1 \\ 2 & 1 & 0 & 4 \\ 2 & 0 & 1 & 1 \\ 1 & 3 & 5 & 0 \end{bmatrix}$$

exiting the recursion when $n = 2$, i.e., computing the products of 2×2 matrices by the brute-force algorithm.

8. Given an $n \times m$ chocolate bar, you need to break it into nm 1×1 pieces. You can break a bar only in a straight line, and only one bar can be broken at a time. Design a divide-and-conquer algorithm that solves the problem with the minimum number of bar breaks.

9. A tromino (more accurately, a right tromino) is an L-shaped tile formed by three 1×1 squares. The problem is to cover any $2^n \times 2^n$ chessboard with a missing square with trominoes. Trominoes can be oriented in an arbitrary way, but they should cover all the squares of the board except the missing one exactly and with no overlaps.



Design a divide-and-conquer algorithm for this problem.

10. Devise an $O(n \log n)$ -time algorithm that takes a sequence S of n integers $a_1, \dots, a_n$ and a separate integer x as input, and outputs 'YES' if there are two integers in S such that the sum of them is x and 'NO' otherwise. To get full credit, you need to provide the efficiency analysis.

11. You are given two sorted sequences which consist of distinct elements. Design an algorithm that finds the K -th smallest element in the union of both sequences. The running time of your algorithm should be $O(\log n + \log m)$ in the worst case where n and m are the sizes of the sequences.

DYNAMIC PROGRAMMING

- 1.** Devise a dynamic programming algorithm that finds the number of different ways to climb an n -stair staircase if each step is either one or two stairs. For example, a 3-stair staircase can be climbed three ways: 1-1-1, 1-2, and 2-1.
- (a) Define a subproblem.
 - (b) Construct the recurrence relation.
 - (c) Determine base cases.

2. Given a string $x = x_1 x_2 \dots x_n$ of length n , design a dynamic programming that checks whether the string x a palindrome or not. A palindrome is a sequence of characters which read the same backward as forward (KABAK, ABBA, ...)

- (a) Define a subproblem.
- (b) Construct the recurrence relation.
- (c) Determine the base cases.

3. Given a string $x = x_1 x_2 \dots x_n$ of length n , design a dynamic programming that finds the minimum number of characters to be inserted to make the string x a palindrome. A palindrome is a sequence of characters which read the same backward as forward (KABAK, ABBA, ...)

- (a) Define a subproblem.
- (b) Construct the recurrence relation.
- (c) Determine the base cases.

4. Suppose you are given a shelf of books, and your job is to divide them among k workers so that they scan the books to find some codes. You can divide the shelf into k regions and assign each region to a worker. (The books ordered according to the number of pages) What will be the fairest way to divide the book among k workers? (the total number of pages each worker gets will be close to each other)

- (a) Define a subproblem.
- (b) Construct the recurrence relation.
- (c) Determine base cases.

5. Devise a dynamic programming algorithm that takes three bit strings

$$X = x_1 \dots x_n$$

$$Y = y_1 \dots y_m$$

$$Z = z_1 \dots z_{n+m}$$

and output 'YES' if Z is an interleaving of X and Y , and 'NO' otherwise. Note that the string Z is called an interleaving of X and Y if Z can be obtained by interleaving the bits in X and Y in a way that it maintains the left-to-right order of the bits in X and Y .

For example, if $X = 101$ and $Y = 01$, then $Z_1 = 10011$ is an interleaving of X and Y , but $Z_2 = 11010$ is not.

6. Devise a dynamic programming algorithm that takes a sequence S of n integers $a_1, \dots, a_n$ as input, and outputs the longest and strictly increasing subsequence. Note that the numbers in the subsequence don't need to be successive.

7. Given a sequence of n houses $x = x_1, \dots, x_n$ where x_i represents the value of the i -th house. Design a dynamic programming that finds the maximum amount of money that can be robbed. The constraint is that two adjacent houses cannot be robbed simultaneously.

- Define a subproblem.
- Construct the recurrence relation.
- Determine base cases.
- Fill in the dynamic programming table given below.

i	1	2	3	4	5
x_i	7	4	11	2	9
OPT(i)					

8. A positive integer can always be represented as a sum of squares of other positive integers. Note that 1 is a square and a positive integer is always written as $(1^2 + 1^2 + \dots + 1^2)$. Given a positive integer N , devise a dynamic programming that finds the minimum number of squares that sum to N . For instance the integer 100 can be written as $(1^2 + 1^2 + \dots + 1^2)$, $\dots$, $(5^2 + 5^2 + 5^2 + 5^2)$, $\dots$, (10^2) . Your algorithm should output 1 for the input 100 since it can be written as (10^2) .

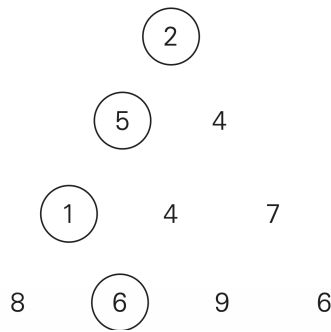
- Define a subproblem.
- Construct the recurrence relation.
- Determine the base cases.

9. Solve the following problem using dynamic programming.

item	weight	value	capacity $W = 6$.
1	3	\$25	
2	2	\$20	
3	1	\$15	
4	4	\$40	
5	5	\$50	

- Apply the bottom-up dynamic programming algorithm to the instance given above of the *Knapsack* problem.
- How many different optimal subsets does the instance of part (a) have?
- In general, how can we use the table generated by the dynamic programming algorithm to tell whether there is more than one optimal subset for the Knapsack problem's instance?

10. Some positive integers are arranged in an equilateral triangle with n numbers in its base like the one shown in the figure below for $n = 4$. The problem is to find the smallest sum in a descent from the triangle apex to its base through a sequence of adjacent numbers (shown in the figure by the circles). Design a dynamic programming algorithm for this problem and indicate its time efficiency.



11. Several coins are placed in cells of an $n \times m$ board, no more than one coin per cell. A robot, located in the upper left cell of the board, needs to collect as many of the coins as possible and bring them to the bottom right cell. On each step, the robot can move either one cell to the right or one cell down from its current location. When the robot visits a cell with a coin, it always picks up that coin. Design a dynamic programming that finds the maximum number of coins the robot can collect and a path it needs to follow to do this.

- Define a subproblem.
- Construct the recurrence relation.
- Determine base cases.

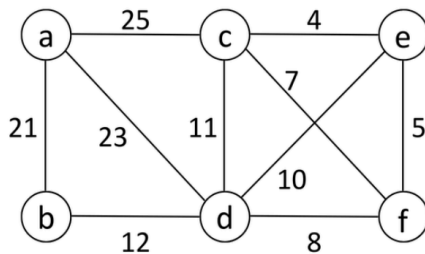
12. Design a dynamic programming algorithm for the following problem. Find the maximum total sale price that can be obtained by cutting a rod of n units long into integer-length pieces if the sale price of a piece i units long is p_i for $i = 1, 2, \dots, n$. What are the time and space efficiencies of your algorithm?

GREEDY ALGORITHMS

1. Given are N ropes of different lengths, devise a greedy algorithm that connects these ropes into one rope with minimum cost. The cost to connect two ropes is calculated as the sum of their lengths. To get full credit, you need to provide the proof for the correctness of the algorithm. For the Input $[5, 2, 4, 8]$. Assume you first connect 5 and 2, and get a rope of length 7. Then you connect 7 and 4, and get a rope of length 11. Finally, you connect 11 and 8, and get a rope of length 19. Then, the total cost will be $7 + 11 + 19 = 37$.

2. Given a row of n coins with the values $v_1, \dots, v_n$ where n is even, consider a game played by two players who alternate turns. In each turn, a player must select either the first or last coin from the row, remove it from the row permanently, and collect the value of the coin. The game continues until no coins remain, and the player with the highest total value wins. Design a greedy algorithm that determines the maximum possible amount of money a player can guarantee to win if he/she moves first. Provide an example that the greedy algorithm you have developed fails to achieve the optimal result.

3. Consider the undirected, weighted graph shown below.



You are running Prim's algorithm to obtain the minimum spanning tree of the given graph. During the execution of the algorithm, what are the intermediate values of the key and the parent parameters for the node C and E.

4. Given an array of size n that has the following specifications:

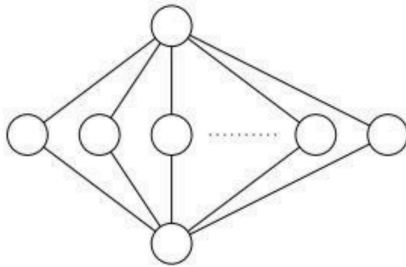
- each element in the array contains either a policeman or a thief,
- each policeman can catch only one thief,
- a policeman cannot catch a thief who is more than K units away from the policeman.

For the array [P, T, T, P, T] and $k = 1$, your algorithm should output 2

For the array [T, T, P, P, T, P] and $k = 2$, your algorithm should output 3.

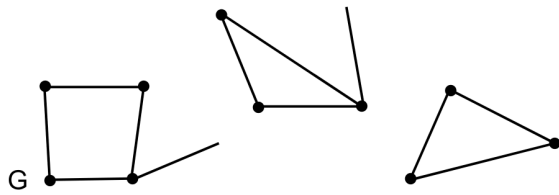
- Devise a greedy algorithm that finds the maximum number of thieves that can be caught.
- Prove that your algorithm satisfies the greedy-choice property.

5. Consider an undirected graph which consists of two poles connected through a layer of nodes as follows:



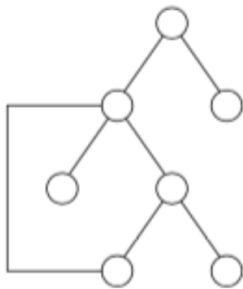
We call such a graph a pole graph. Assume you are given a weighted pole graph consisting of n nodes (each edge has a positive integer weight). Design an efficient algorithm (better than Kruskal and Prim) for finding the minimum spanning tree of the given graph. To get full credit, you need to argue the running time.

6. A connected component of a graph is a set of vertices where each node can reach every other node in the component along the given edges, and which is connected to no additional vertices. For example, the graph G below has 3 connected components.



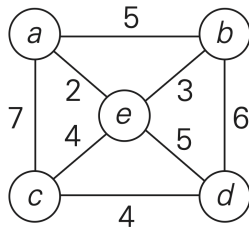
Explain how to use Kruskal's algorithm to compute the number of connected components in a graph.

7. Consider an undirected graph which is formed by taking a binary tree and adding an edge from exactly one of the leaves to another node in the tree. We call such a graph a loop graph. An example of a loop graph could be the following one:

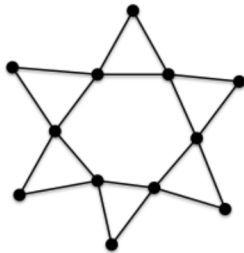


Assume you are given a weighted loop graph (each edge has a positive integer weight). Design an efficient algorithm (better than Kruskal and Prim) for finding the minimum spanning tree of the given graph. To get full credit, you need to argue the running time.

8. Apply Prim's algorithm to the following graph and draw the minimum spanning tree (MST) upon the termination of the algorithm.



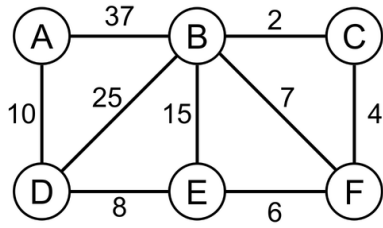
9. A star graph is an undirected graph of the following form, i.e. let n be the number of vertices, a star graph consists of a center cycle with $n/2$ vertices where each pair of connected vertices are also connected to a third vertex:



Assume you are given a weighted star graph (each edge has a positive integer weight).

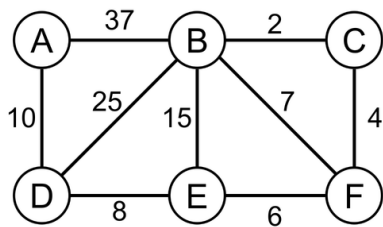
- How long would it take Prim's or Kruskal's algorithms to find the minimum spanning tree of a given star graph (as a function of n)?
- Design an efficient algorithm (better than Kruskal and Prim) for finding the minimum spanning tree of the given graph. To get full credit, you need to argue the running time.

10. Consider the undirected, weighted graph shown below.



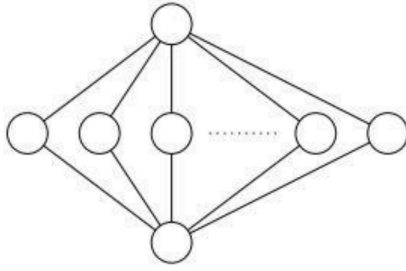
You are running Dijkstra's algorithm to compute the shortest paths from the source node A to all other nodes in the graph. During the execution of the algorithm, what are the intermediate values of the key and the parent parameters for the node B?

11. Consider the undirected, weighted graph shown below.



Draw the minimum spanning tree output by Prim's algorithm using the node A as the source vertex. To get full credit, you need to show the keys of the nodes at each tree.

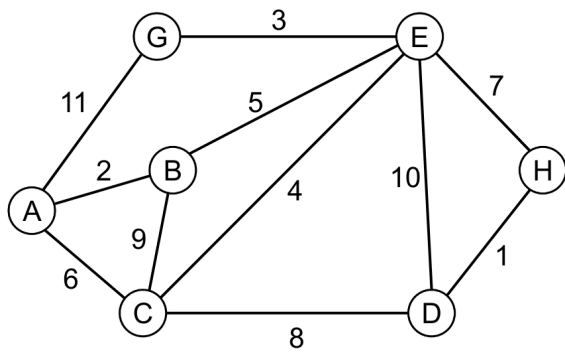
12. Consider a pole-graph which is an undirected graph with positive edge weights, consisting of two poles connected through a layer of nodes as follows:



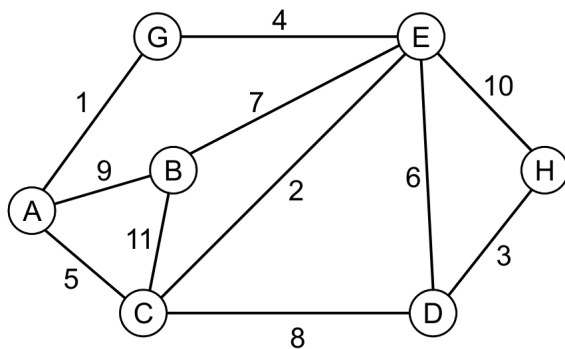
(a) How long would it take Dijkstra's algorithm to find the single-source-shortest-paths from one of the poles in a pole-graph to all other nodes (as a function of n where n is the number of vertices in a pole-graph)?

(b) Describe and analyze a more efficient algorithm for solving the single-source-shortest-paths (SSSP) problem on a pole-graph. To get full credit, you need to argue the running time.

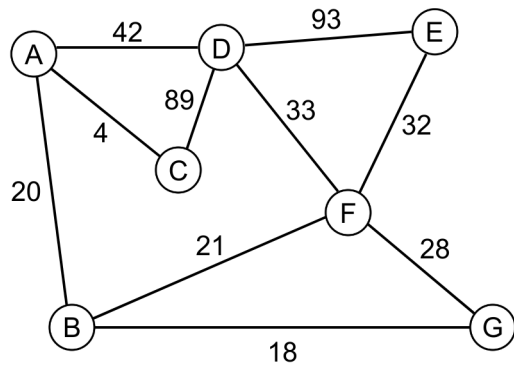
13. For the following graph, use Prim's minimum spanning tree (MST) and Dijkstra's shortest path algorithms. Draw the minimum spanning tree output by Prim's, and the shortest path tree output by Dijkstra's. To get full credit, you need to show the keys of the nodes at each tree.



14. For the following graph, use Prim's minimum spanning tree (MST) and Dijkstra's shortest path algorithms. Draw the minimum spanning tree output by Prim's, and the shortest path tree output by Dijkstra's. To get full credit, you need to show the keys of the nodes at each tree.



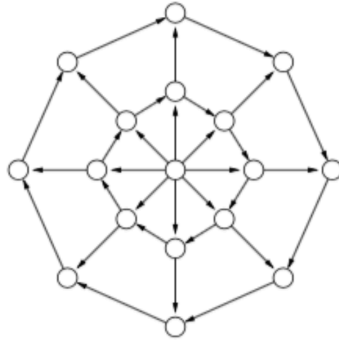
15. Consider the undirected, weighted graph shown below.



(a) Apply Prim's algorithm using the vertex "A" as the source vertex. For the answer, just draw the MST tree that you obtain after the algorithm terminates (to get full credit, you should reflect the final keys of the vertices to the graph).

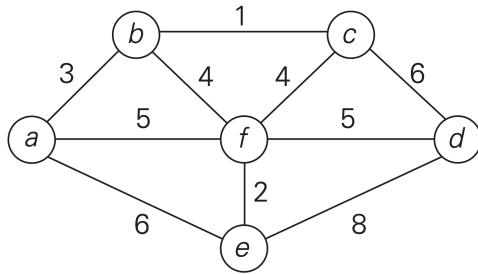
(b) Apply Dijkstra's algorithm using the vertex "A" as the source vertex. For the answer, just draw the shortest path tree that you obtain after the algorithm terminates (to get full credit, you should reflect the final keys of the vertices to the graph).

16. A wheel graph is a directed graph of the following form, i.e. a wheel graph consists of a center vertex c with k outgoing 'spokes' of s outward oriented edges at each circle; furthermore, all the spokes at each circle are connected to form a directed cycle, and all cycles are oriented the same way. ($k = 8$ and $s = 2$ for the following figure)



- (a) How long does it take for Dijkstra's algorithm to output the shortest paths from center to all other nodes in a given wheel graph (as a function of n where n is the number of vertices in the graph)?
- (b) Design an efficient algorithm (better than Dijkstra) for finding the shortest paths from center to all other nodes in a given wheel graph. To get full credit, you need to argue the running time.

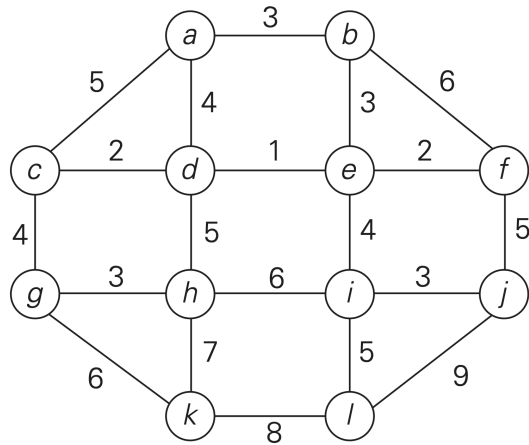
17. Consider the following undirected, weighted graph:



(a) Apply Prim's algorithm using the vertex "a" as the source vertex. For the answer, just draw the MST tree that you obtain after the algorithm terminates (to get full credit, you should reflect the final keys of the vertices to the graph).

(b) Apply Kruskal's algorithm using the vertex "a" as the source vertex. For the answer, just draw the MST tree that you obtain after the algorithm terminates (to get full credit, you should reflect the final keys of the vertices to the graph).

18. Consider the following undirected, weighted graph:



(a) Apply Prim's algorithm using the vertex "a" as the source vertex. For the answer, just draw the MST tree that you obtain after the algorithm terminates (to get full credit, you should reflect the final keys of the vertices to the graph).

(b) Apply Kruskal's algorithm using the vertex "a" as the source vertex. For the answer, just draw the MST tree that you obtain after the algorithm terminates (to get full credit, you should reflect the final keys of the vertices to the graph).

19. Consider the following string:

"EEKMEKSUEKMEKSUEKMEKSUEKMEKMS".

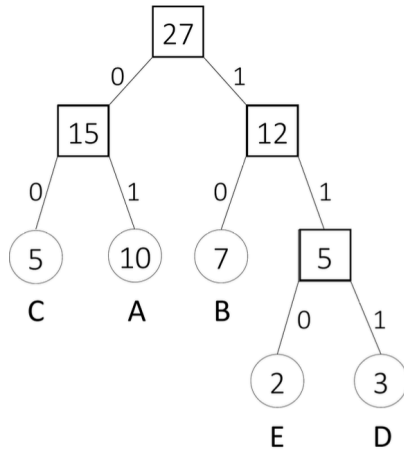
- (a) Find the frequencies of the characters in the string, and construct the Huffman tree corresponding to these frequencies.
- (b) Based on the Huffman tree you have obtained, determine the Huffman codes of the characters, and calculate the cost of the encoding for your string.

20. Consider the following string:

"ABBACCADABRAABBACCADABRARABBACCADABRAABBACCADABRA".

- (a) Find the frequencies of the characters in the string, and construct the Huffman tree corresponding to these frequencies.
- (b) Based on the Huffman tree you have obtained, determine the Huffman codes of the characters, and calculate the cost of the encoding for your string.

21. A prefix tree is provided for character frequencies of a string.



(a) Calculate the total cost of the encoding of the string considering the given tree.

(b) Construct a Huffman tree using the same string, then calculate the total cost and compare it with the prefix cost.

CRYPTOGRAPHY

1. For an RSA public key $(N, e) = (55, 7)$ and a message $m = 2$, calculate the ciphertext of the message m . Also, break the encryption scheme and calculate the secret key.
2. For an RSA public key $(N, e) = (77, 9)$ and a message $m = 2$, calculate the ciphertext of the message m . Also, break the encryption scheme and calculate the secret key.
3. A polyalphabetic cipher such as the *Vigenere* cipher is stronger against letter frequency analysis when compared to a monoalphabetic cipher like *Caesar*. Explain why?